

BrickAtlas Display-Grounded Evidence Following Scientific Supplement

Anonymous CVPR submission

Paper ID Draft

1. Reading Guide and Version Boundary

This supplement documents the display-v2 observation contract, human qualification, response statistics and graph audit. The full 617 historical question dossiers are distributed separately as `dossiers.json` and a searchable HTML interface. They are not 617 new display-v2 questions. Historical prompts and answers remain historical even when a parent ID is linked to new observations.

The analysis version is `ldraw2-evidence-v2`; the visual dataset is `brickatlas-display-v2`. Canonical source identity is the source hash and placed-instance ID. Display references resolve within one current image. The complete four-part identity rule and all parent-to-observation mappings are released with the dataset. No old model answer or review is transferred to a new prompt.

The artifact has three distinct inspection modes. The evaluator site exposes pairing, gold and provenance for scientific audit. The blind review packages expose only one exact observation at a time, without gold or pair membership. The historical 3D viewer allows rotation, zoom, source-part isolation, expansion and visibility replay. Only the blind packages implement the qualification protocol; richer 3D access cannot substitute for judging native model-input pixels.

2. Exact Visual Stimuli

2.1. Reference questions and a concrete pair

For the OMR 42004 Color example, both arms ask:

What color is the object currently marked B0035 in this image?

Both include the rule that references select current image objects and that appearances or labels may have been edited. Gold changes from Blue (choice D) to Red (choice C). Original source truth stores Black. This is an explicit rendered-appearance counterfactual, not a claim that the source record changed.

For the corresponding Part-type example, both arms ask:

Which displayed candidate label currently marks an object with the same visible part geometry as the target marked B0035? Ignore color and pose. Answer from this image, not from historical source identities.

Candidate B0036 in A has the same source type, 32250, as the target. After the B0036/B0196 candidate-reference swap, B0196 selects that same matching geometry in B. The option-ID gold changes from C to A. This describes generated golds only. There is no observed model C-to-A or C-to-C response case in the current release.

Figure 1 in the main paper uses the actual native PNGs with an identical crop rectangle for the two arms within each family. Its annotations are paper-only. `figure-provenance.json` records the four observation IDs, full PNG hashes and crop rectangles. Full images and full prompts remain available through the paired-observation inspector.

2.2. Rendering audit

All primary and auxiliary images have a native resolution of 1280×800 . Color edits keep the geometry, pose, camera, label projections, visibility and material settings identical. The body mask is the projected part body; one pixel of dilation accommodates antialiasing at the silhouette. Changed pixels outside that mask number zero in every Color pair. The mask test is not a measurement of human color separability, projected difficulty or assembly stability.

The 73 pairs use two distinct named opaque colors each; both arms may differ from the historical source appearance. Aggregate category counts are 37 Red, 37 Blue, 36 Yellow and 36 Black. Canonical color, part type and transforms are preserved separately. The donor feasibility inventory retains every candidate considered for the older mechanism and explains why a general donor swap cannot be relabeled as a color-only intervention.

Part-type image pixels are inherited from the preserved operand-only and candidate-swap renders; their prompts and observation identities are new. Independent gold validation reconstructs the current display mapping and checks

source part-type equality. Whether that equality is uniquely inferable visually is deliberately left to human review.

2.3. Position-reference control

The 67 auxiliary observations each have a target panel and four candidate panels, without labels inside the image (Figure 1). The question describes which panel is the target and the ordering of the four candidates. The correct option is derived independently from source geometry/type correspondence. No candidate-reference change pair is manufactured from these panels.

Each auxiliary observation requires two independent raw human reviews, for 134 additional judgments. Its final decidable set and accuracy are separate from the 280 primary endpoints and their paired Both. Comparing it with a label-based arm changes several factors at once: reference interface, separation of objects and view access. A higher score would not isolate geometry skill, and a lower score would not isolate position understanding.

3. Human Review Protocol

3.1. Sampling universe and blinding

Both reviewer packages contain exactly 347 units: 146 Color endpoints, 134 Part-type endpoints, and 67 position controls. Each has its own opaque unit IDs and shuffled order. Deduplication cannot use the parent ID, because one parent has multiple distinct reviewed observations. The binding includes exact text and PNG bytes, not merely an image path.

Reviewers work independently before model inference and do not receive pair membership, condition labels, canonical source records, gold, model responses or another person's answers. The coordinator distributes one package to each of two different people. The native image is scrollable rather than silently downsampled. A reviewer attests that the full native observation was inspected.

3.2. Raw and final records

Every raw record contains:

- opaque unit ID and exact wire-observation hash;
- reviewer pseudonym, timestamp and independence attestation;
- decidable, ambiguous, unanswerable or contradictory decision;
- chosen visual option where applicable;
- four Boolean reference/answer/visibility/contradiction checks;
- reason code, free-text reason and native-size attestation.

A decidable record requires a listed option, all four checks passing, and the no-exclusion reason code. Nondecidable records require a reason. Raw records are never overwritten by a final decision. Duplicate raw reviews, reuse of a

reviewer identity across the two slots, obsolete hashes and synthetic evidence are rejected.

If categorical decisions, choices and checks agree, the record is finalized under the frozen rule. Different explanatory prose alone does not constitute a categorical disagreement. Any decision, choice or check disagreement requires a third person's adjudication. The adjudicator sees the single observation and the two judgments, but not gold, the other endpoint or model responses. A reason is required. A final decidable choice conflicting with the generated gold returns the item to pending and requires dataset revision.

3.3. Agreement and gate semantics

For doubly reviewed units, raw decision agreement is the fraction of equal categorical decisions. Multiclass Cohen's kappa is

$$\kappa = (p_o - p_e) / (1 - p_e), \quad (1)$$

where p_e is computed from the two empirical decision marginals. No doubly reviewed units yields null; degenerate marginals with $p_e = 1$ also yield null. Choice agreement and each structured check have separate statistics. Primary and auxiliary counts are explicit. Adjudicated agreement does not replace raw agreement.

Before real inference, every planned endpoint needs a final nonpending decision. This gate does not require every endpoint to be decidable: ambiguous, unanswerable and contradictory outcomes remain part of the full benchmark report. Additional QA-subset estimates require both endpoints to be finally decidable. Missing either A or B blocks that pair's inclusion. The final summary is reconstructed from retained raw inputs; editing a summary cannot satisfy the gate.

At release, raw human reviews are 0/560 for primary endpoints and 0/134 for auxiliary controls. Final primary decisions are 0/280; both-decidable pairs are 0/140. These are unresolved acceptance requirements. Software tests exercise the rule with temporary synthetic fixtures; none is imported as a real review.

4. Statistics and Genuine Response Evidence

4.1. Outcomes and denominators

For a changed-answer pair, B outcomes are mutually exclusive: correct new answer, valid old answer, another valid answer, or invalid. Their micro proportions sum to one over all planned pairs. A accuracy and Both additionally distinguish the A endpoint. Missing calls do not remove the pair. JSON validity is reported for each arm and for both, not inferred from semantic correctness alone.

Conditional adaptation uses only A-correct pairs. Micro adaptation is the both-correct count divided by that count. Source-macro adaptation first computes each source's conditional rate, then averages over sources with at least one



Figure 1. Complete position-reference control image for the OMR 42004 example. The upper panel is the target and the four lower panels are candidates referenced by ordinal position in the question. No text is embedded in the image. The correct candidate is third from the left among the four candidates. These isolated source parts are not a proposed assembly. This interface changes layout and visible access, so its contrast with candidate labels is descriptive, not an identified OCR effect.

A-correct pair. The report includes both eligible pair and eligible source counts. Those two estimands can differ; neither replaces unconditional Both.

For invariance graph pairs, $y_A = y_B$. Both remains the joint success criterion. A repeated wrong answer is consistent but incorrect. Stability of raw invalid strings has no semantic meaning. The current visual release does not import older camera/ID-renaming invariance responses under the new display contract.

4.2. Sources, subset shifts and rank comparisons

Source-macro estimates give equal weight to sources after averaging their within-family pairs. Whole-source resampling uses 10,000 draws and a fixed seed, 20260920. Paired model differences use the same items and resampling weights. The QA-minus-full calculation recomputes both means using shared source draws and the subset’s actual source coverage. A draw containing no eligible subset source is recorded as undefined; it is not assigned a zero score.

Rank points are computed using exact rational source means. Equal values receive midranks, avoiding arbitrary floating-point thresholds for ties. The rank comparison reports A, NewAcc and Both, the sign of each paired difference, point reversals versus ties, descriptive intervals and

agreement of directions across full/QA sets. With many families or models, these unadjusted comparisons remain exploratory. An interval crossing zero does not prove equivalence.

4.3. Response cases and authenticity boundary

Only completed live runs validated against the new request contract can enter visual analysis. Each model must use one common finalized QA snapshot. The runner records raw response bytes, extracted output text, normalized choice, model revision and full-request hash. Analysis reparses raw responses; it does not trust manually entered accuracy columns. Missing or malformed responses remain failures.

Case extraction uses genuine QA-decidable pairs, separately by family and model. It looks for A-correct to new gold, A-correct to old gold, and the same valid wrong answer at both endpoints. Selection is deterministic by hash order within a case type. An absent type stays absent, and its absence is not filled with a constructed response. At release there are no eligible real cases and no empirical rank comparison. The example in Section 2 describes stimulus semantics only.

The collection pipeline checks hashes, provenance and declared response models. Local files and attestations alone cannot cryptographically prove that a remote provider gen-

erated a receipt or that two humans worked independently. Retained provider records and coordinator oversight complement the machine checks.

5. Graph Partitions and Algorithmic Results

5.1. Full strong-pair inventory

Table 1 lists all 12 strong changed-answer pairs. Every row matches surviving nodes, edge count, target degree, complete degree multiset and isolate count. BFS and union-find are both correct on every row. Anonymous degree and 1-WL feature predictors are jointly incorrect on every row because their restricted features do not distinguish the two golds.

Table 1. Every strong change pair; 12 pairs from eight sources. The final column is the exact per-pair result, not an average over generated variants. BFS and union-find have Both=1 on every row. The machine audit additionally supplies full feature signatures, source-held-out lookup support and predictions.

Parent task	y_A	y_B	Isolates+1 Both
ld2-10001-graph-removal-4	2	3	0
ld2-10014-graph-removal-1	2	3	0
ld2-10014-graph-removal-2	1	2	0
ld2-10213-graph-removal-1	4	3	0
ld2-31009-graph-removal-2	4	3	0
ld2-5867-graph-removal-1	1	2	0
ld2-omr-10269-graph-removal-1	2	1	0
ld2-omr-10269-graph-removal-2	2	1	0
ld2-omr-10269-graph-removal-4	2	1	0
ld2-omr-21309-graph-removal-2	2	3	0
ld2-omr-21309-graph-removal-4	2	1	0
ld2-omr-42054-graph-removal-4	3	4	0

Table 2. Natural graph source-macro accuracy in percent, with descriptive whole-source percentile intervals. All 73 items and 24 sources are retained. Lookup training excludes the test source and is separate from hypothetical graphs.

Algorithm	Source macro	95% interval
Node-count lookup	83.75	[73.12, 92.92]
Edge-count lookup	61.11	[46.39, 75.21]
Degree-histogram lookup	93.96	[87.71, 99.17]
Anonymous 1-WL lookup	93.96	[87.71, 99.17]
Isolates + one	97.92	[93.75, 100.00]
BFS	100.00	[100.00, 100.00]
Union-find	100.00	[100.00, 100.00]

5.2. Natural distribution and invariance

Table 2 supplies natural source intervals; the main paper reports exact item counts. A node-count *lookup* is distinct from the older hard-coded $|V| - 1$ predictor: it learns a

feature-to-answer table on other sources. The distinction is preserved in filenames, labels and reports.

Table 3 gives both invariance partitions. The 73 anchors are reused from the change pairs, so counting all pair endpoints would yield 292 appearances but only 219 distinct observations. The lookup training population includes changed endpoints once and excludes isomorphic invariance copies. This prevents those copies from disproportionately weighting a source label.

Table 3. Both-correct on invariance pairs. Strong and degree-visible sources overlap with the corresponding changed-answer partitions. Stable predictions need not be correct.

Algorithm	Strong / 12	Degree-visible / 61
Node-count lookup	6	31
Edge-count lookup	7	32
Degree-histogram lookup	6	61
Anonymous 1-WL lookup	6	61
Isolates + one	6	61
BFS	12	61
Union-find	12	61

5.3. Answer balance and small-sample limits

The strong direction count is six increases and six decreases. Table 4 discloses that raw arm answer frequencies are not identical. Both is invariant to swapping endpoint roles. Symmetric endpoint accuracy averages the two endpoints within each existing pair. Optional equal arm weighting uses each endpoint in both arm roles with half weight; it changes neither observations nor pair/source counts. It does not create uniform label frequencies.

Table 4. Stored strong-change answer histograms. Equal numbers of increases/decreases do not imply equal per-answer arm frequencies.

Stored arm	Gold 1	Gold 2	Gold 3	Gold 4
A	2	7	1	2
B	4	2	5	1

The Wilson intervals $[0, .2425]$ and $[.7575, 1]$ for 0/12 and 12/12 are approximate because source reuse violates pair independence. Boundary whole-source bootstraps may collapse to a point. The released intervals and per-pair data expose these limitations rather than imply a precisely known population effect.

All strong graphs instantiate a narrow construction: six-cycle versus two triangles, plus identical isolates. Uniform-feature anonymous 1-WL cannot distinguish them at any refinement depth. A degree or 1-WL lookup cannot evade this collision by fitting labels more aggressively. A model

with unique-ID features, positions, spectral information or higher-order computation is outside this feature restriction. No trained GNN or language-model result is claimed by the algorithmic audit.

claims must contract accordingly. No receipt-free score, invented rank reversal or synthetic human judgment can close these requirements.

317
318
319

6. Source Coverage and Historical Controls

Table 5 makes source coverage explicit. Source files come with recorded URLs, author attribution, license metadata and content hashes in the historical catalog. Dataset-level paired analysis cannot substitute for source-level split discipline. Any future training/evaluation split must keep each source and all its derived views, interventions and controls together. Common part libraries may still create dependencies across such splits.

The historical families remain an inspection resource. Distance reads coordinates and computes a value; Interface and Coverage read supplied records; Step lookup reads author annotations. Neighbors and Graph deletion operate the explicitly supplied graph. Evidence limit and Step sequence include fixed-answer controls. Restoration follows a visibility contract at a saved source pose. None establishes physical insertion planning.

The viewer can replay these actions, expose the original graph/records and inspect part IDs. Expansion is an inspection transformation, not a physically valid final arrangement. Isolation omits supports without changing the saved assembly. Source-local overlap candidates still require physical review. The new color and label interventions do not claim to have corrected the historical source geometry.

7. Reproduction and Acceptance Checklist

The source package contains this manuscript, its main paper, generated tables, figures, bibliography and the CVPR style. Both PDFs compile with Tectonic. The repository additionally contains the publication generator, frozen stimuli, review packages, validators and raw-input hashes. The build guide specifies the checked runtime and commands.

The following machine checks are complete in this release: 347 frozen native visual observations; 140 paired gold changes; 73 controlled body-pixel differences; all visual golds independently recomputed; 219 graph request snapshots and dual-algorithm gold checks; and 26 protocol/analysis regression tests. Browser and PDF inspection validate delivery behavior and layout. Those inspections are not human visual-task certification.

Empirical acceptance remains open for the 560 primary human reviews, 134 auxiliary reviews and any necessary adjudications, followed by new model inference. The required analysis must show full and QA-decidable results, valid-format rates, family-specific response transitions, paired source intervals and genuine case examples where they exist. If the findings fail to reveal added diagnostic value, the

Table 5. All source assemblies. Band is source scale. Color and Part-type count primary parents, each contributing exactly one A/B pair. All 24 sources appear in both visual families; variants do not increase source count.

Source	Name	Band	Parts	Color	Part-type
31028	Sea Plane	D1	53	2	2
31027	Blue Racer	D1	67	2	2
10156	LEGO Truck	D1	111	2	1
omr-42102	Mini CLAAS XERION	D1	129	2	1
omr-42020	Twin-Rotor Helicopter	D1	141	2	1
10036	Pizza To Go	D2	166	2	2
10014	Caboose	D2	170	2	2
omr-31031	Rainforest Animals	D2	215	2	2
omr-31088	Deep Sea Creatures	D2	230	2	1
omr-42004	Mini Backhoe Loader	D2	246	2	2
omr-42061	Telehandler	D2	261	2	2
31009	Small Cottage	D2	270	2	1
5867	Super Speedster	D2	278	2	2
10128	Train Level Crossing	D2	336	2	2
10159	City Airport	D3	592	3	3
10001	Metroliner	D3	848	4	4
omr-10269	Harley-Davidson Fat Boy	D4	1029	4	4
omr-10242	MINI Cooper	D4	1086	4	4
omr-42066	Air Race Jet	D4	1244	5	5
10213	Shuttle Adventure	D4	1254	5	5
omr-10265	Ford Mustang	D4	1357	5	5
10220	Volkswagen T1 Camper Van	D4	1428	5	4
omr-21309	NASA Apollo Saturn V	D4	1845	5	5
omr-42054	CLAAS XERION 5000 TRAC VC	D4	1978	5	5

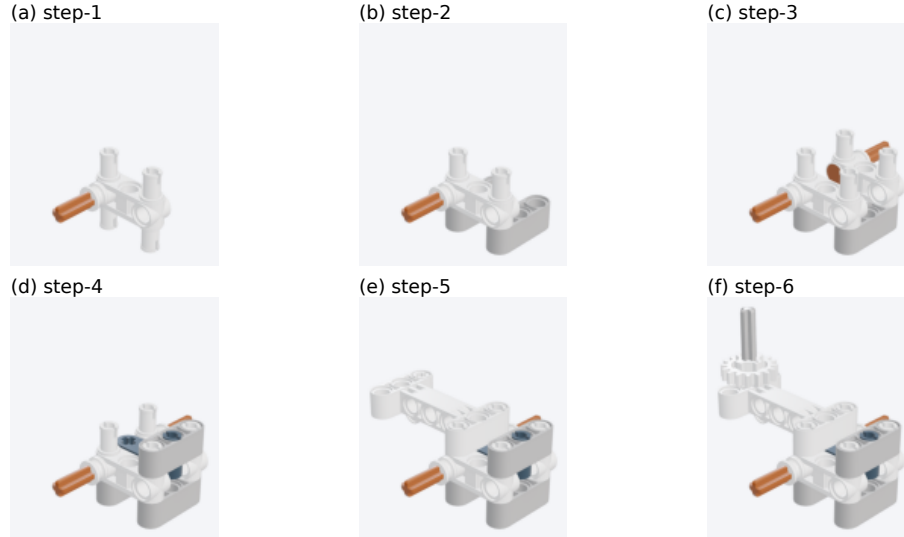


Figure 2. Historical given-order visibility replay for set 42102, retained as a 3D inspection example. Six accepted actions reveal parts at original source transforms. The sequence is supplied by the control, not discovered by a model. Display correspondence does not demonstrate a collision-free insertion path or gravity stability. The figure is reused with its original file hash recorded in the new publication provenance.